Group 5 — Simulated Annealing for TSP

Cover fields to complete: member names, IDs, contributions, and private repository URL.
Instances: att48, ch130, a280

1. Natural inspiration | 2 minutes

Annealing heats metal to allow particle movement, then cools slowly to settle into a stable structu

SA mapping:
- High T: exploration; accept some worse tours.
- Low T: exploitation; focus on improvements.
- Cooling: gradually reduces disruption.

2. Mathematical formulation | 3 minutes

Tour = permutation of city indices.

Δ = f(candidate) − f(current)
Accept if $\Delta \leq 0$; otherwise accept with $P=\exp(-\Delta/T)$.
T_new=T_old·r, r=(T_final/T_initial)^(1/blocks).
2-opt replaces (a,b),(c,d) with (a,c),(b,d).

Numerical example

Current cost = 120; candidate cost = 130; $\Delta=10$; $T=20$.
$P=\exp(-10/20)=0.6065$.
Random draw 0.40 → accept.
Random draw 0.80 → reject.

Initial T is estimated from typical worsening 2-opt moves and the target acceptance.

3. Benchmark protocol | 4 minutes

TSPLIB: att48 (48, ATT, BKS 10628), ch130 (130, EUC_2D, BKS 6110), a280 (280, EUC_2D, BKS 25

30 runs per method, seeds 2026–2055, 50,000 objective evaluations per run.
Baseline: random-restart improving 2-opt.

Results

SA mean gaps: att48 2.59%, ch130 9.04%, a280 46.65%.
Baseline mean gaps: att48 14.26%, ch130 42.52%, a280 76.00%.

Mann–Whitney p-values: approximately 3.0×10^{-11} for every instance; SA medians are significantly

Convergence and sensitivity

Show the convergence and final-cost plots from results/plots/.
Show the six tuning heat maps from results/plots/tuning/.

Selected settings: att48 (0.7, 0.75, 0.1, 200), ch130 (0.3, 0.25, 0.5, 50), a280 (0.1, 0.25, 0.5, 200)

4. Synthesis | 1 minute + Q&A

Strength: accepts temporary worsening and escapes local optima.
Sensitivity: initial temperature and cooling schedule matter.
Limitation: fixed budget and simple 2-opt neighborhood scale poorly; a280 retains a 46.65% mear

References: Kirkpatrick et al. (1983), Metropolis et al. (1953), Reinelt (1991), Croes (1958).